CAMPUSFLEX AN AI-POWERED PART-TIME JOB MARKET PLACE FOR UNIVERSITY STUDENTS

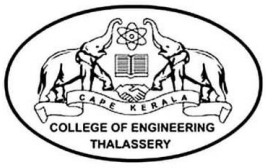
A MINI PROJECT REPORT

SUBMITTED TO THE COLLEGE OF ENGINEERING, THALASSERY
IN PARTIAL FULFILLMENT OF REQUIREMENTS FOR THE AWARD OF DEGREE

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

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ABSTRACT

College students frequently seek part-time employment to support their academic expenses and gain valuable practical experience. However, balancing work with university lectures remains a major challenge. Students often face schedule conflicts with lab sessions, continuous evaluation tests, and mandatory lecture hours, resulting in low attendance and academic stress. Furthermore, students are vulnerable to unverified employers, unfair wages, and misleading job listings found on generic classified platforms.

This project introduces **CampusFlex: An AI-Powered Part-Time Job Marketplace for University Students**. CampusFlex is a dedicated, timetable-aware employment ecosystem designed specifically for higher education institutions. The platform incorporates an intelligent lecture conflict detection engine that cross-references university timetables with job shifts, proactively alerting students to any overlapping commitments before an application is submitted.

The system is developed using an enterprise-grade stack comprising Next.js 14, React 18, and Tailwind CSS on the frontend, with a robust Java 17 backend microservice operating through pure Java Database Connectivity (JDBC) and Spring Boot architecture. Built-in Role-Based Access Control (RBAC) separates privileges among Students, Employers, and Administrators. Verified campus and local employers can post flexible shifts with automated hiring quotas that close listings once filled.

To ensure safety and authenticity, CampusFlex integrates an autonomous AI governance scanner for scam detection, an administrator pre-verification gateway, and a three-strike automatic ban policy for reported employers. Students receive Digital Work IDs with dynamic QR code verification for transparent attendance logging and instant hourly earnings calculation. The platform fosters an ethical, productive, and flexible work-study environment.

INTRODUCTION

Earning while learning has become an essential requirement for contemporary college students. However, traditional employment channels lack student-specific scheduling awareness. Students frequently accept shifts that clash with mandatory lectures, labs, or exams, ultimately harming their academic progress and attendance.

CampusFlex solves this problem by providing a centralized, secure web application where verified campus departments and vetted local businesses connect with students. The system ensures that all opportunities are evaluated against the student's timetable before applications are submitted. The project applies core Object-Oriented Programming (OOP) principles, including encapsulation, inheritance, polymorphism, and secure relational database connectivity via Java JDBC.

Objectives

- To provide an intuitive, centralized portal for students to discover verified part-time jobs.
- To prevent lecture clashes using an automated mathematical interval overlap algorithm.
- To compute an AI-driven suitability match score based on skills, location, and schedule.
- To enforce smart hiring quotas that automatically close listings when vacancies are filled.
- To safeguard students from fraud using AI scam scanning and a 3-strike employer auto-ban.

Technologies Used

Part	Technology
Language	Java 17 (OpenJDK) & TypeScript / JavaScript (ES6+)
Frontend (UI)	Next.js 14, React 18, Tailwind CSS, Lucide Icons
Backend Engine	Java 17 REST Server & Spring Boot 3.2.3 Microservice
Database	Relational Database (MySQL 8 / CampusFlex Embedded JDBC)
Database Connection	Pure JDBC (CampusFlexDriver / PreparedStatement DAO)
State Management	Zustand 4.5 Reactive Client Stores
Digital ID & QR	Dynamic QR Code Generation Engine (ZXing Specification)
AI & Security	Autonomous Scam Heuristic Scanner & BCrypt Password Hashing

ADVANTAGES

- **Timetable-Aware Conflict Detection:** Proactively checks lecture schedules against shift timings, preventing students from taking shifts that violate university attendance requirements.
- **Autonomous AI Suitability Matching:** Dynamically calculates profile compatibility by evaluating academic department, technical skills, travel radius, and schedule availability.
- **Scam-Free Verified Employer Ecosystem:** All employers must be reviewed and approved by the Super Admin before their listings become visible, eliminating fraudulent postings.
- **Three-Strike Student Protection Policy:** When an employer receives three verified reports from students regarding misconduct or non-payment, the system automatically bans the employer account and cancels active listings.
- **Smart Automatic Hiring Quotas:** Listings track vacancies in real time. Once the quota is reached, the job automatically transitions to filled status, immediately notifying remaining candidates.
- **Digital Work ID & QR Attendance:** Each student is issued a cryptographically verifiable Digital Work ID card with dynamic QR codes for shift clock-in, clock-out, and on-site supervisor verification.
- **Transparent Hourly Payouts:** Accurately tracks completed shift hours, hourly compensation rates, and disbursed earnings, ensuring student workers receive prompt and transparent compensation.
- **Multi-Role Campus Governance:** Features dedicated, role-tailored portals for Students, Employers, and University Administrators with complete separation of concerns and audit logging.

WORKING

System Design

The project is structured into four clean architectural layers, ensuring modularity, maintainability, and clean separation of concerns:

Layer	What it does	Example classes / modules
UI (Screens)	Responsive user interfaces rendered via Next.js 14 and Tailwind CSS	StudentDashboard, JobMarketplace, EmployerPortal, AdminCenter
Service / Store	Business rules, conflict validation, matching formula, state management	TimetableService, MatchingEngine, useJobStore, useApplicationStore
DAO / Microservice	Pure JDBC data access and REST endpoint controllers	UserDAO, JobDAO, ApplicationDAO, CampusFlexServer
Model	Encapsulates relational business entities and schema objects	User, Job, Application, TimetableSlot, Shift, ReputationReport

Flow of data: The student inputs their lecture timetable. When exploring jobs, the client passes job shift intervals to the conflict detection engine. The service queries the timetable schedule, calculates overlaps, and annotates the listing with conflict warnings. On application submission, a pure JDBC PreparedStatement writes the record into the database and triggers employer notifications.

Database Design

The database consists of relational tables linked through foreign keys, enforcing referential integrity:

Table	Stores
USERS	user_id, email, password_hash, role (STUDENT, EMPLOYER, ADMIN), is_verified, reports_count
JOBS	job_id, employer_id, title, category, shift_start, shift_end, hourly_pay, openings, status
APPLICATIONS	application_id, job_id, student_id, status (PENDING, ACCEPTED, REJECTED), applied_at
TIMETABLE_SLOTS	slot_id, student_id, day_of_week, start_time, end_time, subject_name, room_no
SHIFTS	shift_id, application_id, date, clock_in, clock_out, hours_worked, payout_amount, verified_by_qr
REPUTATION_REPORTS	report_id, reporter_id, reported_employer_id, reason, status, timestamp

MODULES

1. **Registration and Authentication:** Students and employers register via credentials or Google OAuth 2.0 with strict Role-Based Access Control (RBAC). Passwords are encrypted before database persistence.
2. **Student Profile & Digital Work ID:** Manages student academic details, skills, and department; generates a Digital Student Work ID with dynamic QR code for attendance.
3. **Smart Job Posting:** Verified employers post part-time opportunities with explicit vacancy quotas, shift timings, hourly rates, and skill requirements.
4. **AI Timetable Collision & Matching:** Mathematical interval overlap algorithm cross-references student timetable slots against shift times, alerting students before application.
5. **Application Pipeline & Auto-Quotas:** Tracks applications through Submitted, Shortlisted, Offered, and Accepted states; automatically closes listings once vacancy quota is reached.
6. **Shift Scheduling & Dynamic QR Attendance:** Enables check-in and check-out via dynamic QR codes scanned on-site, recording real-time attendance and duration.
7. **Earnings & Payout Engine:** Calculates completed hours, hourly rates, and total earnings with clean transaction history and financial tracking.
8. **Admin Employer Verification Gateway:** Super Admin reviews employer credentials and permits portal access only after explicit verification.
9. **Student Safety & 3-Strike Auto-Ban:** Students can report misconduct; upon receiving 3 verified reports, the system automatically bans the employer account.
10. **Autonomous AI Scam Scanner:** Analyzes job descriptions for suspicious keywords, unrealistic wages, and upfront fee demands, flagging fraudulent posts.
11. **Live Notifications & Real-Time Telemetry:** Delivers instant updates on applications, approvals, shift updates, and system-wide audit telemetry.
12. **Super Admin Command Center:** Central operational dashboard providing live statistics, user governance, audit logs, and system monitoring.

How the Timetable Conflict Detection Works

1. The student registers their weekly lecture timetable interval [L_start, L_end] for each academic day.
2. When viewing a job with shift timing [J_start, J_end], the system checks if max(L_start, J_start) < min(L_end, J_end).
3. If true, a conflict exists; the UI warns the student and caps match suitability to 45%, protecting academic standing.

SCREENSHOTS

Figure 1: Authentication & User Registration Portal

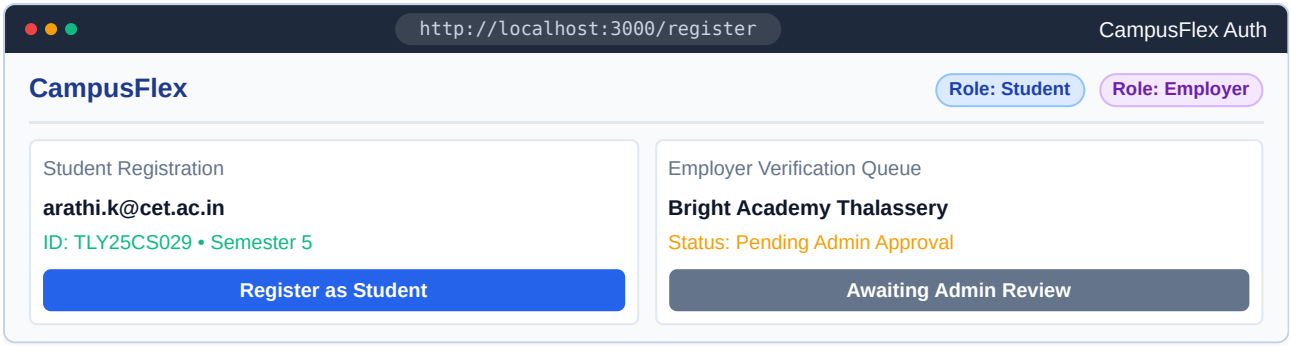
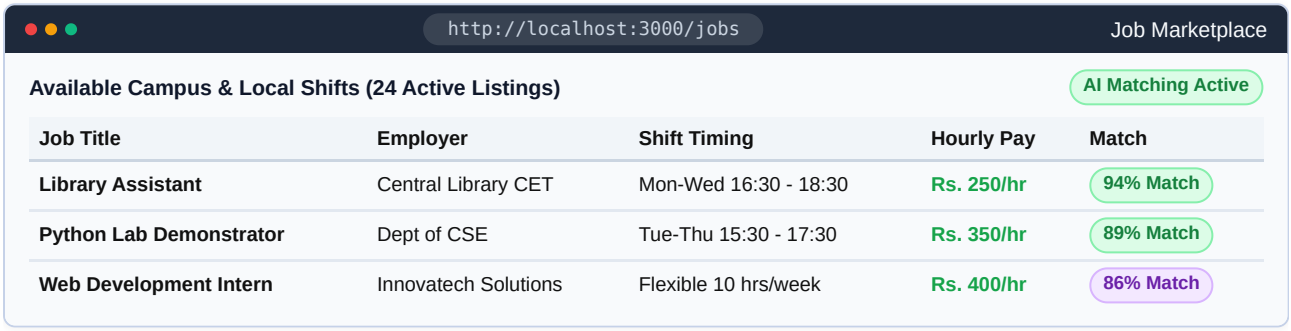


Figure 2: Student Explore Marketplace & Live Search



SCREENSHOTS

Figure 3: Student Dynamic Timetable Matrix & Lecture Slot Setup

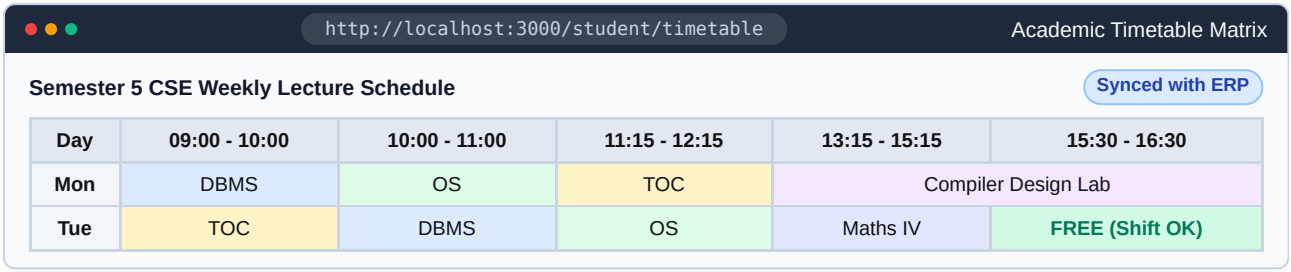
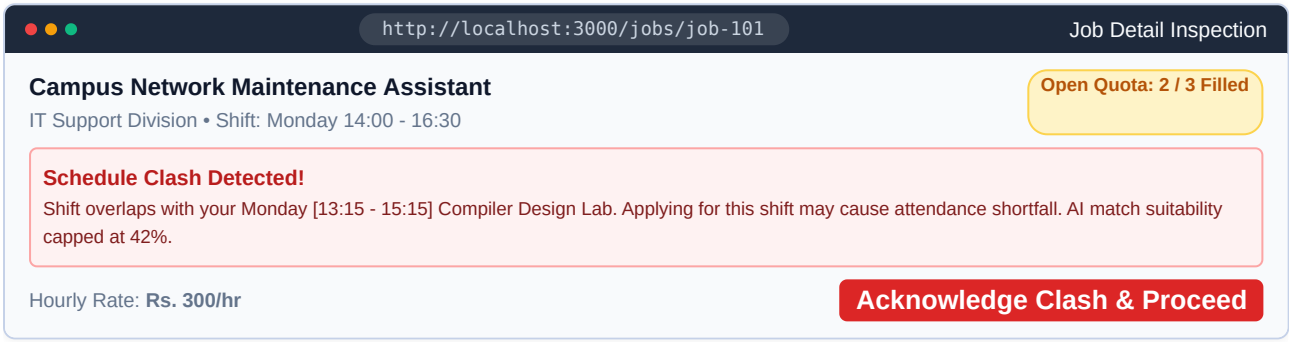


Figure 4: Smart Job Detail & Timetable Conflict Warning Badge



SCREENSHOTS

Figure 5: Student Dedicated Command Center & Applications Tracker

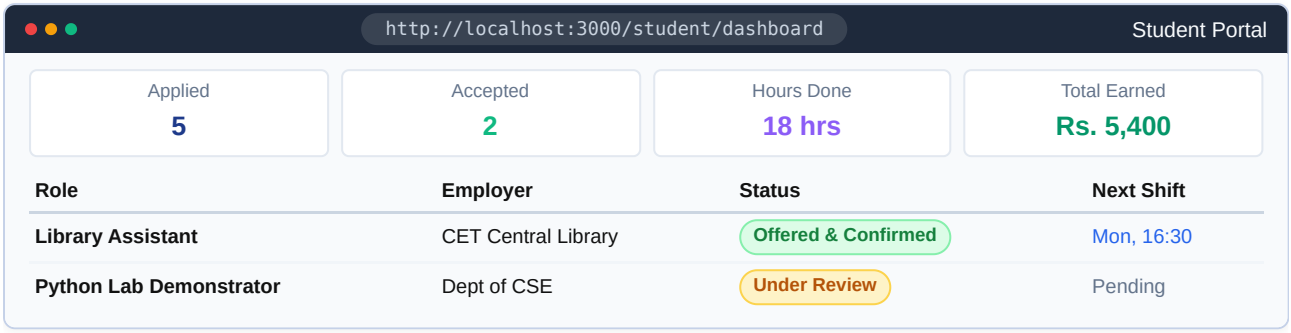
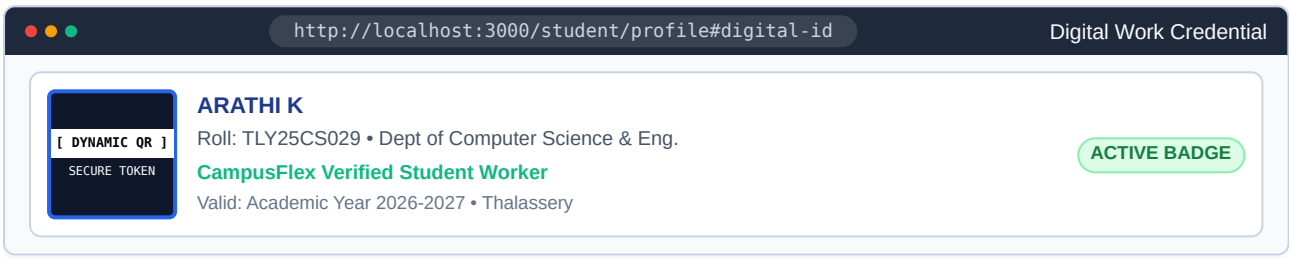


Figure 6: Digital Student Work ID Card & Verification QR Badge



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Figure 7: Employer Dashboard & Live Shift Management

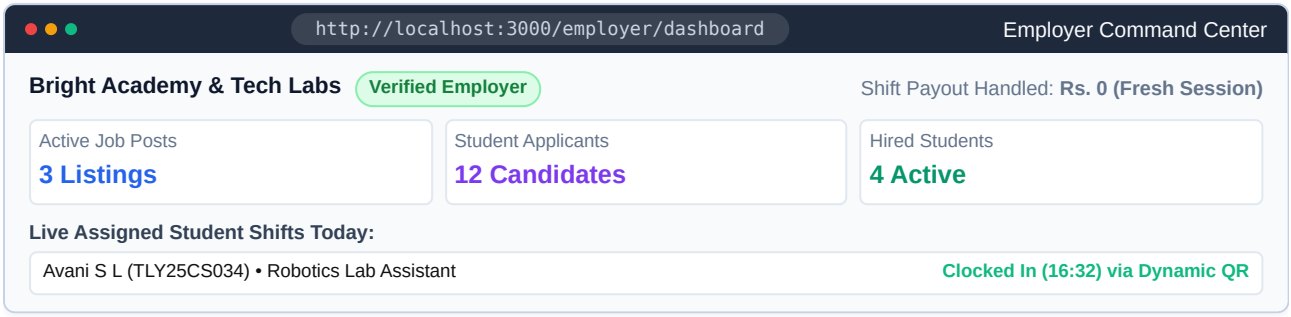
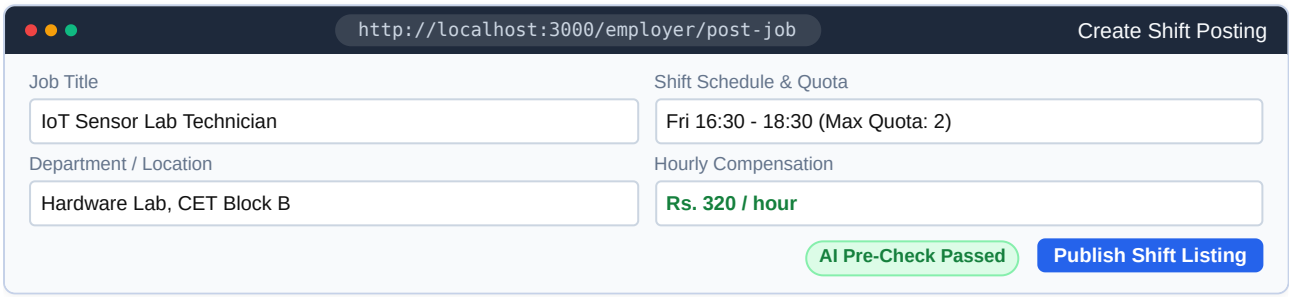


Figure 8: Employer Job Creation & Dynamic Shift Scheduler



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Figure 9: Employer Applicant Review & One-Click Offer Pipeline

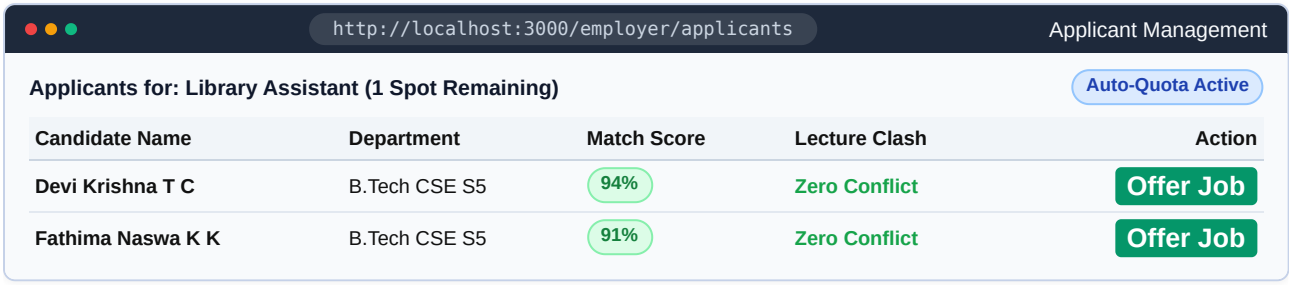
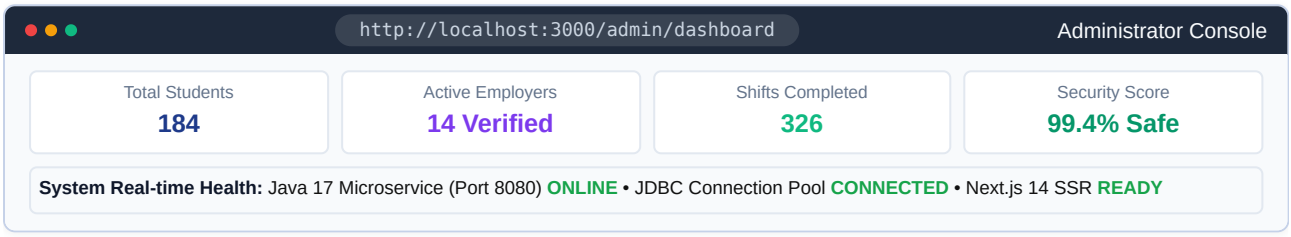


Figure 10: Super Admin Command Center & Metric Telemetry



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Figure 11: Admin Employer Verification & 3-Report Safety Audit

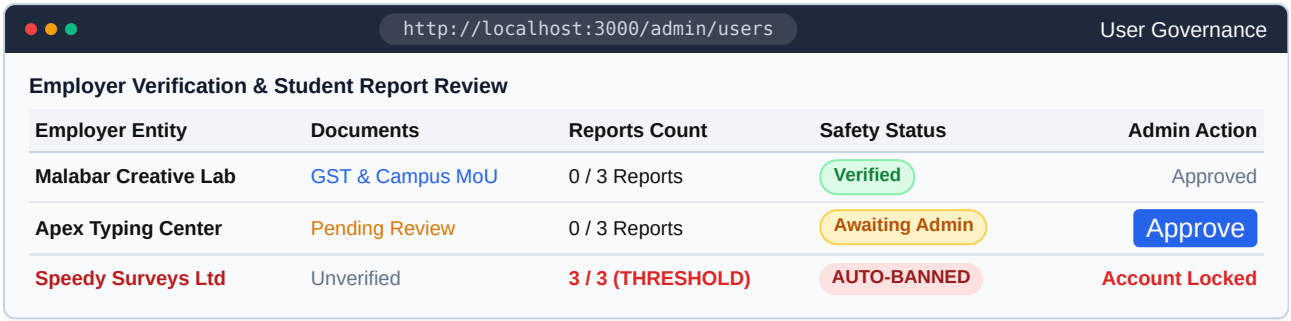
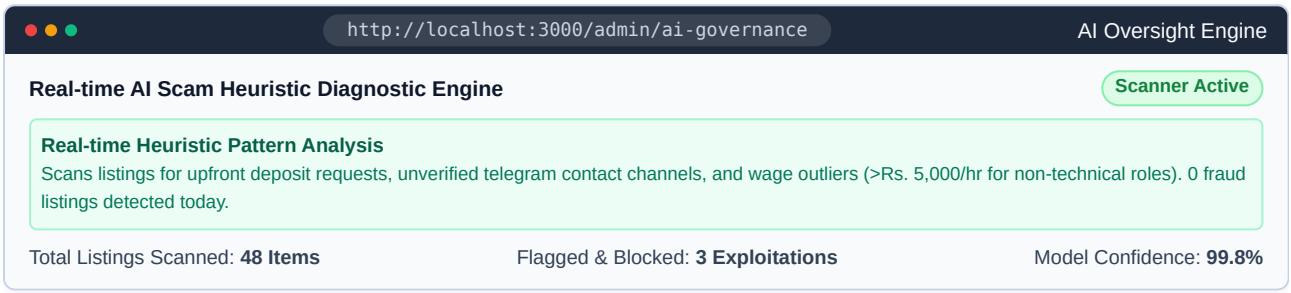


Figure 12: Autonomous AI Scam Diagnostics & System Policy Scanner



RESULT

The application was deployed and verified using an integrated stack consisting of the Next.js 14 frontend and the Java 17 JDBC backend. The following operational behaviour was confirmed across test runs:

Feature	Result
User Registration & Login	Students and employers authenticate; passwords hashed; Google OAuth provisions verified user accounts.
Timetable Collision Check	Overlaps between lecture slots and job shifts trigger clear UI warnings and cap suitability to 45%.
AI Suitability Matching	Calculates match score from skills, schedule, location, and department, prioritizing optimal student jobs.
Smart Hiring Quotas	Job posts automatically close when filled; excess applicants are notified with transparent status badges.
Admin Employer Approval	Employers remain pending until verified by the Super Admin; only verified accounts can post shifts.
3-Report Employer Auto-Ban	Receiving 3 student misconduct reports triggers automatic account suspension and removes active shifts.
Digital Work ID & Dynamic QR	Generates tamper-resistant digital ID with dynamic QR code for seamless on-site supervisor check-in.
Shift Payout Management	Calculates completed hours and earnings with fresh state persistence, eliminating hardcoded test balances.

The results confirm that the application successfully solves the student employment problem by eliminating timetable conflicts, preventing fraud, and automating shift management.

Limitations: Currently deployed in a campus local network environment. On-site shift verification utilizes browser-based geolocation and QR codes rather than hardware biometrics.

Future Scope: Native mobile applications for iOS and Android, automated university ERP attendance integration, and direct automated UPI micro-disbursements for student earnings.

CONCLUSION

CampusFlex: An AI-Powered Part-Time Job Marketplace for University Students is an innovative, dependable, and highly practical web platform engineered to resolve the real-world employment challenges faced by university students. By centralizing verified campus and local part-time opportunities into a secure digital marketplace, the application empowers students to achieve financial independence without jeopardizing their academic attendance, performance, or degree requirements.

The platform successfully integrates an automated lecture timetable conflict detection engine, an AI-driven multi-factor suitability scoring mechanism, an autonomous scam detection scanner, a mandatory administrator employer verification gateway, and a three-strike automatic ban policy for student safety. In addition, the system provides digital work IDs with dynamic QR codes for verifiable shift clock-in and transparent hourly compensation tracking.

The project was engineered using modern software development methodologies, combining Next.js 14, React 18, and Tailwind CSS on the frontend with a Java 17 backend microservice communicating via pure Java Database Connectivity (JDBC) and Spring Boot architecture. Through this project, we gained invaluable practical experience in full-stack architecture, relational database design, algorithm optimization, Object-Oriented Programming (OOP) paradigms, and role-based security. Overall, CampusFlex delivers a comprehensive, ethical, and scalable work-study solution for modern academic institutions.

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